Accepted: 24 February 2017

DOI: 10.1111/sms.12868

ORIGINAL ARTICLE

WILEY

Velocity-specific and time-dependent adaptations following a standardized Nordic Hamstring Exercise training

T. Alt[1] | Y. T. Nodler[1] | J. Severin[1] | A. J. Knicker[1,2] | H. K. Strüder[1]

[1]Institute of Movement and Neuroscience, German Sport University, Cologne, North Rhine-Westphalia, Germany

[2]SPRINZ Sport Performance Research Institute of New Zealand, University of Technology, Auckland, New Zealand

Correspondence
Tobias Alt, Institute of Movement and Neuroscience, German Sport University, Cologne, North Rhine-Westphalia, Germany.
Email: t.alt@dshs-koeln.de

The Nordic Hamstring Exercise (NHE) is effective for selective hamstring strengthening to improve muscle balance between knee flexors and extensors. The purpose of this study (within subject design of repeated measures) was to determine the effects of a standardized 4-week NHE training on thigh strength and muscle balance with concomitant kinetic and kinematic monitoring. Sixteen male sprinters (22 years, 181 cm, 76 kg) performed a standardized 4-week NHE training consisting of three sessions per week (each 3×3 repetitions). Six rope-assisted and six unassisted sessions were performed targeting at a constant knee extension angular velocity of ~15°/s across a ~90-100° knee joint range of motion. Kinetic (peak and mean moment, impulse) and kinematic parameters (eg, ROM to downward acceleration, ROM_{DWA}) were recorded during selected sessions. Unilateral isokinetic tests of concentric and eccentric knee flexors and extensors quantified muscle group-, contraction mode-, and velocity-specific training adaptations. Peak moments and contractional work demonstrated strong interactions of time with muscle group, contraction modes, and angular velocities (η^2>.150). NHE training increased eccentric hamstring strength by 6%-14% as well as thigh muscle balance with biggest adaptations at 150°/s 2 weeks after NHE training. Throughout the training period significant increases (P<.001) of peak (η^2=.828) and mean moments (η^2=.611) became apparent, whereas the impulse and the ROM_{DWA} of unassisted NHE repetitions remained unchanged (P>.05). A 4-week NHE training significantly strengthened the hamstrings and improved muscle balance between knee flexors and extensors. Despite the slow training velocity, biggest adaptations emerged at the highest velocity 2 weeks after training ended.

KEYWORDS
eccentric resistance training, injury prevention, knee joint, muscle balance, muscle strength ratio

1 | INTRODUCTION

Since the beginning of this century, the Nordic Hamstring Exercise (NHE) has gained attention in sports sciences.[1,2] It has been identified as an effective exercise for improving thigh muscle balance because it highly activates the hamstrings without involving the quadriceps.[3] This exercise incorporates a controlled lowering of the trunk and thighs from a kneeling position while maintaining an extended hip which forces the hamstrings to work eccentrically. Since 2002, NHE training has received significant research attention for the first time[1] and has been implemented in different sports (eg, soccer, rugby, Australian rules football). The eccentric mode of NHE is superior to concentric exercise in promoting strength gains[4] and more effective compared to a traditional hamstring curl exercise regimen.[5] Despite its bilateral nature, impairing its practical transferability to running,[6] the NHE stresses both limbs equally,[7] reduces bilateral strength

© 2017 John Wiley & Sons A/S. Published by John Wiley & Sons Ltd

asymmetries,[8] and improves agonist-antagonist muscle ratios (H_{ecc}:Q_{con}) in uninjured participants.[5,9]

Several large cohort cluster randomized controlled trials (n=114-958) consistently reported the positive effect of NHE training on hamstring strain injury (HSI) rates.[10-15] However, a recent survey in professional soccer teams revealed that NHE implementation in regular training schedule is generally low.[16] This finding was remarkable as the NHE is an inherent part of the FIFA 11+ warm-up procedure and NHE training compliance is crucial for injury prevention.[17] Studies with substantially smaller cohorts (n=8-15) investigated the effects of NHE training on isokinetic parameters[5,7-9,18,19] and reported increased hamstring peak moments of 2%-21%. Due to differences in training period (4-10 weeks), volume (9-30 sessions with a total of 160-720 NHEs), participants (from adolescents to professional athletes), and isokinetic test conditions (eg, concentric or eccentric testing), a reasonable recommendation for purposeful NHE training can hardly be derived.

Since 2013, a NHE device enabled the reliable quantification of NHE training load and progress.[20] This device allows the scientific monitoring and evaluation of large cohorts.[21-24] These impressive data sets supply the strongest evidence that eccentric hamstring strength is associated with HSI risk. It is not yet sufficiently established if recently introduced body mass dependent target values of peak moments improve the comprehensible rating of hamstring strength and enhance HSI prevention.[25] However, muscle architectural properties such as the fascicle length also seem to play an important role in HSI prevention.[24]

So far training studies have not verified NHE execution by kinematic analyses. Two studies executed 2D motion captures during single NHE training sessions.[26,27] The angle at the instant of downward acceleration (DWA angle) was identified as an important characteristic of NHE performance representing the lost control of lowering the trunk with at a constant knee extension velocity. DWA angles displayed a significant relationship to hamstrings eccentric peak moment (R^2=.65) revealing that NHE can be applied as a field-based assessment tool.[28] Delahunt et al.[29] investigated the effects of a 6-week partner-assisted NHE training on NHE kinematics and isokinetic parameters. The training significantly improved hamstrings eccentric peak moments (15%) and DWA angles (8%).

Despite this extensive research on NHE over the past 15 years, a systematic and comprehensive kinematic and kinetic analysis of standardized NHE training is still lacking. This might be due to a limited number of technical opportunities. It has been assumed that NHE cannot be performed on a dynamometer.[26] The recording of NHE kinetics is currently limited to a single device.[20] Another limitation is that NHE execution has usually been based on target velocities indicated as average cadence (eg, 30°/s). Consequently, the time under tension near full knee extension was frequently very short.[26,27,29] This might have diminished the training progress as this specific range of motion (ROM) (30-0° knee flexion) provokes a long hamstring fascicle length which is presumably crucial for effective HSI prevention.[6,24] Avoiding excessive extension torques by assisting the trunk's lowering with a decelerating rope during early NHE training stages might prevent this deficiency.

Thus, the aims of the present study were (a) to conduct a standardized 4-week NHE training on an isometrically operating dynamometer, (b) to include rope-assisted "isokinetic" training sessions performed with a constant knee extension velocity as well as (c) to quantify the training progress by kinetic and kinematic NHE data and (d) isokinetic parameters.

2 | MATERIALS AND METHODS

2.1 | Participants

Sixteen regional to national class male sprinters (age: 21.6±2.5 years, body height: 181.4±7.1 cm, body mass: 75.7±9.8 kg) gave their written informed consent to voluntarily take part in the study. They differed in training history (2-17 years), training volume (4-12 h/wk), and performance level (100 m season's best: 10.99-12.90 s). All were familiar with lower extremity resistance training, but not with NHE. They had neither thigh muscle nor knee injuries within the last 3 years. The local ethics commission confirmed that the requirements of the Declaration of Helsinki were met. During the intervention period all participants maintained their normal physical activity. Thirteen participants performed all 108 NHEs, whereas three omitted one NHE session due to personal reasons. During the 10-week study period all participants maintained their physical fitness so that there was no time loss due to injuries or muscle soreness.

2.2 | Instruments

All isokinetic tests and the NHE training sets were executed on the dynamometer IsoMed2000 (D&R Ferstl GmbH, Hemau, Germany) recording raw angle and moment data at 200 Hz. All relevant test parameters were synchronized and derived from the obtained moment-angle curves (isokinetic tests) and moment-time curves (NHE repetitions) by a self-developed software written in C++. Within selected NHE training sessions, the participants' motion was captured by two high-speed cameras (acA640-120gc; Basler AG, Ahrensburg, Germany) at 100 fps (TEMPLO 8.2.358; Contemplas, Kempten, Germany). Kinematic analysis was conducted with Vicon Peak Motus (V10.0.1; New York, New York, USA).

2.3 | Study design

In total, each participant performed six isokinetic tests and twelve NHE training sessions (Figure 1) following the same

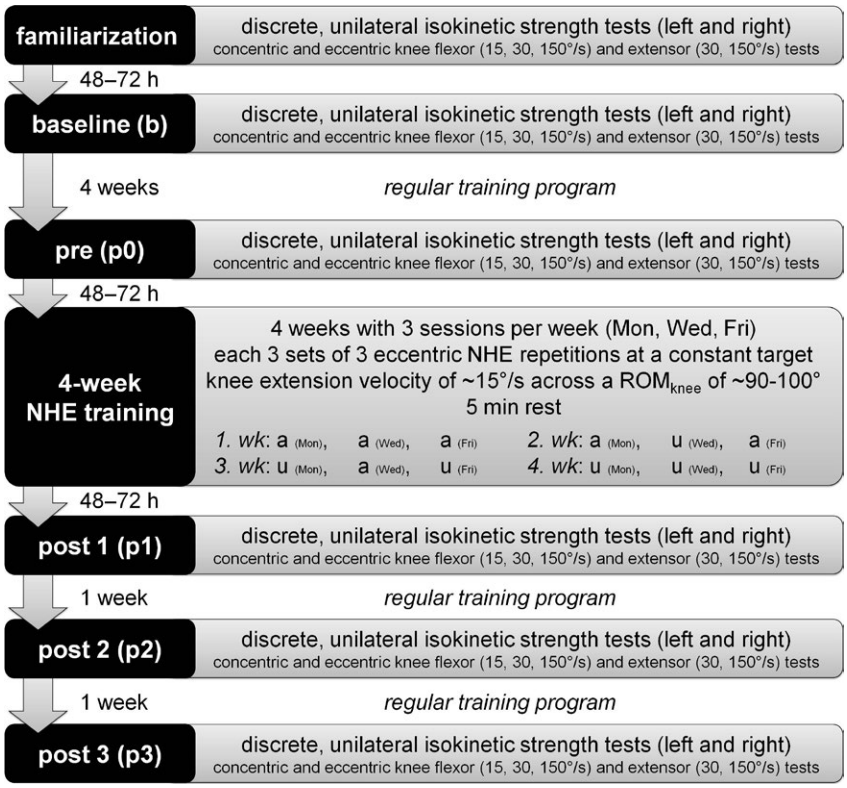

FIGURE 1 Chronological sequence of the 18 sessions containing six isokinetic tests (familiarization, baseline, pre, post 1, post 2, post 3) and twelve NHE training sessions being composed of six assisted (a) and six unassisted (u) sessions

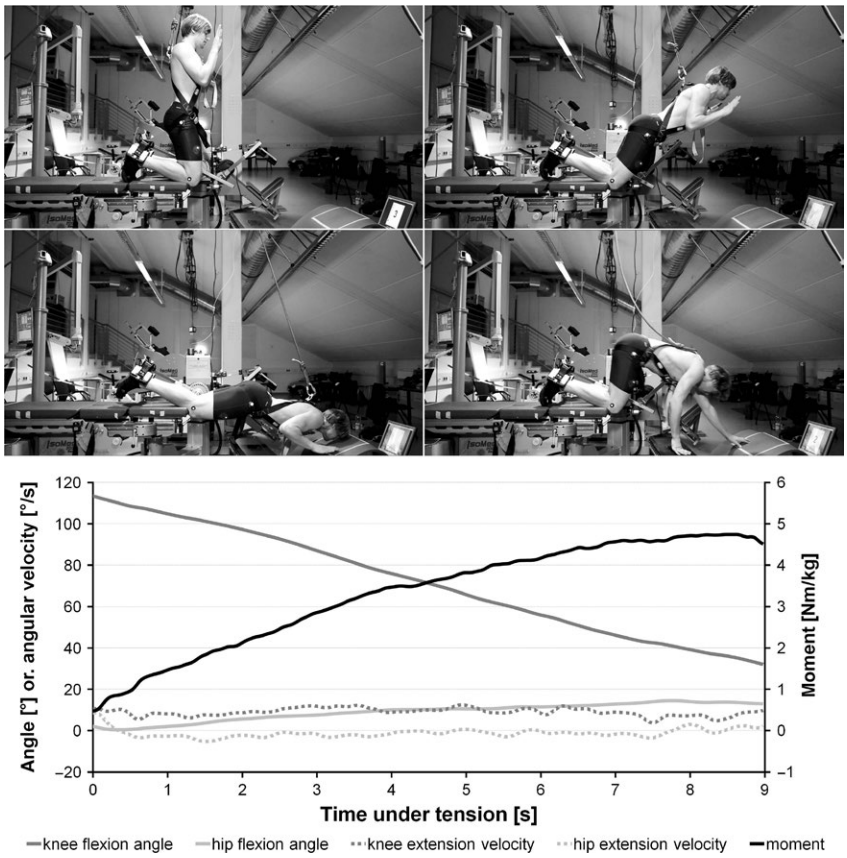

FIGURE 2 Representative illustration of a single rope-assisted NHE repetition (top) including the derived kinetic and kinematic NHE data (bottom). It became apparent that a constant knee extension velocity (9.1°/s) was realized throughout the whole knee joint range of motion (81.5°) whereas the maximum hip flexion angle accounted for 15.4°

initial procedure. After determining their mass, participants performed an individual warm-up (jogging, dynamic stretching) for 10 minutes and rated their DOMS via visual analogue scale (VAS) (0: no pain at all, 10: worst pain ever).[5] The 4-week period between baseline (b) and pretest (p0) represented the baseline effect of their regular training so that

each participant served as his own control subject (within subject design of repeated measures). The 2-week period after NHE training completion (p1-p3) was chosen because this is the usual length of a pre-competitive tapering period. During these 2 weeks all participants conducted their regular training.

2.3.1 | Isokinetic testing procedure

A lever arm with a double shin pad for unilateral knee motion was attached to the motor-driven axis of the dynamometer. The isokinetic tests followed a protocol with proven reliability.[30] In an isometrically contracted muscular state at 90° knee flexion, the dynamometer axis was aligned with the participants' lateral femoral epicondyle with the help of a laser pointer.[32] The double shin pad was fastened by a strap at the lower leg approximately 2-3 cm proximal to the medial malleolus. A static gravity correction was undertaken with the lever arm in a horizontal position, following which the participants conducted an isokinetic warm-up consisting of 15 submaximal concentric and eccentric repetitions of the respective muscle group. The testing order was stratified so that limbs were alternatingly tested and slow velocities were executed prior to fast ones.[33] For knee flexor tests (110-0° ROM_{knee}), participants were asked to lie prone (extended hips) while pushing the trunk with their hands to the lounger. Knee extensor tests (0-90° ROM_{knee}) were performed in supine position (extended hips) with handgrips providing sufficient stability.[32]

Both concentric and eccentric contractions were executed in a discrete movement at 30°/s and 150°/s. Additionally, eccentric knee flexor tests were performed at 15°/s reflecting the NHE training velocity (Figure 1). Each set consisted of five repetitions (two submaximal, three with maximum effort). The last three repetitions were selected for further analysis. An interset rest of 1 min ensured sufficient recovery.[34] Strong verbal encouragement promoted maximum exertion.

2.3.2 | NHE training

All participants kneeled on the lounger so that the knee axis' orientation matched the dynamometer's axis. Prior to each repetition both heels were pushed against two pads attached to the lever arm (Figure 2). The 4-week NHE training consisted of six assisted and six unassisted sessions in which the training modality gradually alternated (Figure 1). The participants were asked to meet the following execution quality criteria: (a) to execute NHEs with a constant knee extension velocity of ~15°/s across the largest possible ROM_{knee} (~90-100°) targeting a time under tension of ~6.5 seconds per repetition and (b) to accomplish the eccentric phase with minimal hip flexion while (c) the hands should be situated close the shoulders. A sagittal live video feedback provided by a webcam recording at 30 fps (C200; Logitech, Apples, Switzerland) and a stick figure moving at target speed were overlaid on a 17′ monitor (710N; SAMSUNG, Seoul, South Korea). Assisted sets were conducted via rope-controlled resistance by the examiner which was transferred to the back part of a climbing harness (Newton; Petzl, Crolles, France) worn by the participants (Figure 2). The examiner adopted the resistance according to the live feedback so that the required knee extension velocity was optimally realized.

After the general warm-up, NHE sessions started with a specific preparation at a doorway pull-up bar (Denqbar DQ-0161, Pirna, Germany) (2×5 repetitions, 3 min rest). These were performed eccentrically at a self-selected velocity until the angle at DWA was reached. The concentric phase was assisted by the examiner. Afterward, 3×3 NHEs (5 min rest) were performed on the dynamometer. Each repetition was executed solely eccentrically while the return to starting position occurred with a push of the arms combined with knee and hip flexion (Figure 2).

During seven of twelve NHE sessions (the last assisted and all six unassisted NHE sessions) the sagittal and transversal motion of the participants were captured by two high-speed cameras. Not all sessions were recorded due to reasons of time required for data processing and analysis. Retroreflective markers (Ø=15-25 mm) were attached on three bony landmarks (acromion, trochanter major, lateral fermoral epicondyle) and on a distinctive position on the lever arm representing the location of the lateral malleolus (Figure 2). Two headlights (Bar Fly 200; Kino Flo® Lightning Systems, Burbank, California, USA) improved the markers' visibility. The measuring volume was statically calibrated with a rigid frame of 912×590×370 mm (L×W×H).

2.4 | Data processing

Raw data of isokinetic and NHE repetitions were stored as ASCII files. For isokinetic analyses, the isokinetic ROM (±1% of the respective angular velocity) was isolated by a self-developed software (C++). A fifth order Butterworth low-pass filter with a cut off frequency of 6 Hz attenuated oscillations of the derived moment-angle curves. For each testing condition, the trial with the highest gravity-corrected peak moment was selected. This measure showed good relative (intraclass correlation coefficient>.83) and absolute (standard error of measurement <8%) reliability scores when participants were familiar with the testing procedure.[30] Additionally, the corresponding angles of peak moment and contractional work were determined. Furthermore, the functional H:Q ratio (H:Q) as well as the moments and angles of the dynamic control ratio at the equilibrium point (DCRe) quantified thigh muscle balance.[35-37] The DCRe represents the intersection point of concentric knee extension and eccentric knee flexion moment-angle curves. Its moment and

angle values possess good reliability and seem to have direct links to hamstring mechanics during the late swing phase of high-speed running.[31] As bilateral differences were beyond the study's scope, all parameter values of both limbs were averaged.

The beginning and the end of a NHE repetition was determined by the first derivative of the moment-time and knee flexion angle-time curves (Figure 2). Peak and mean moment, impulse (M*t), ROM_{DWA}, knee, and hip flexion angles as well as knee extension velocity were selected as relevant NHE parameters. The knee flexion angle at DWA was identified as highest angular acceleration in the knee extension velocity-knee flexion angle curve.[26,29] At the beginning of the NHE training period, each participant chose a knee position either on the lounger, in front of or on the edge of it. This was due to interindividual differences in perception of convenience. Misalignment of external dynamometer and internal knee joint axis was mathematically corrected by calculating the resultant moments for each training session. All isokinetic and NHE data were normalized to body mass allowing inter-and intraindividual comparison.

2.5 | Statistical analysis

For all data, normal distribution and variance homogeneity were confirmed by the Kolmogorov-Smirnov ($\alpha>.10$) and Levene's test ($\alpha>.10$). Two-way and multifactorial ANOVAs with repeated measures identified the effects of NHE training on kinetic and kinematic NHE data (factors: time, execution mode) and on isokinetic parameters (factors: time, muscle group, angular velocity, contraction mode). Bonferroni post hoc tests ($\alpha\leq.05$) determined the actual *P*-values between the five isokinetic tests and the twelve NHE training sessions. All statistical tests were calculated with SPSS (V.23.0; SPSS Inc., Chicago, Illinois, USA). The level of significance was set at $\alpha\leq.05$. Effect sizes are indicated as partial eta-squared (partial η^2) specifying the meaningfulness of the respective results. A partial eta-squared of .02, .13, and .26 represented small, medium, and large effect sizes, respectively.[38]

3 | RESULTS

3.1 | Kinetic and kinematic NHE training progress

All NHE parameters are listed in Table 1. On average, DOMS was generally perceived as mild (2.5±1.5) showing no systematic reduction throughout the training period. Unassisted NHE repetitions resulted in higher peak and mean moments compared to assisted ones (Figure 3A,B). Both parameters showed initial increases which leveled off during week 3 and 4. With regard to the impulse, assisted repetitions accounted for a significantly higher training load (Figure 3C). The impulse generated throughout the first assisted session exceeded the first unassisted session by 38%. This difference was still prominent when comparing to the last respective sessions (21%) (Table 1).

Concerning NHE kinematics, time under tension per repetition was significantly lower during unassisted execution (4.4±1.1 s vs 6.8±0.6 s, $P<.001$, $\eta^2=.732$) (Table 1). The kinematic analysis of the last assisted NHE session (8a) revealed that target values of ROM_{knee} (81.4° vs ~90-100°) and mean knee extension velocity (12.2°/s vs 15°/s) were sufficiently realized. The goal of a constant velocity within assisted NHE execution was met (99.6% ROM_{DWA}). In contrast, unassisted NHE sessions demonstrated mean ROM_{DWA} of 56% (5u) to 80% (10u) ROM_{DWA} (mean of all unassisted NHEs: 68.5±24.4%) revealing no time effect. After the first unassisted NHE training session (5u) showing altered knee kinematics, ROM_{knee} (72.7±8.3°) and mean knee extension velocity (17.3±4.9°/s) did not reveal any training-induced changes. With regard to hip kinematics, maximum hip flexion angles and ROM_{hip} were significantly ($P\leq.05$) smaller during assisted execution. Within the unassisted NHEs, maximum hip flexion (17.0±8.2°) and ROM_{hip} (15.9±6.3°) did not demonstrate any training adaptations (Table 1).

3.2 | Muscle group-, velocity-, contraction mode-and time-dependent adaptations

Consistently, all isokinetic parameters remained unchanged comparing tests at baseline (b) and pre (p0) (Table 2). Peak moments and contractional work revealed strong interactions of time with muscle group, contraction mode, and angular velocity indicating that NHE training selectively improved hamstrings eccentric strength capacities (Table 3). For hamstrings eccentric peak moments, a significant interaction between time and angular velocity became apparent ($P=.007$; $\eta^2=.159$). Biggest adaptations occurred at the highest angular velocity peaking 2 weeks after NHE training completion (p1: +8.5%, p3, +14.3%; $P<.001$; $\eta^2=.552$) (Figure 4A). Hamstrings concentric peak moments significantly increased by 5.6% (30°/s) and 7.7% (150°/s) comparing baseline to post 3 ($P=.001$; $\eta^2=.281$) (Table 2). Strong interactions of time and contraction mode demonstrated that eccentric improvements of hamstrings peak moments ($P<.001$; $\eta^2=.392$) and contractional work ($P<.001$; $\eta^2=.358$) were higher than concentric ones. However, increases of hamstrings eccentric contractional work showed no time dependent interaction with angular velocity (Table 3). Knee flexors' angles of peak moment were unaffected by NHE training just like all isokinetic parameters obtained from the knee extensors (Table 2). Muscle balance expressed as functional H:Q ratio, DCRe moment and angle (Figure 4B,C) revealed significant time effects ($P<.001$) with large effect sizes (Table 2).

TABLE 1 Descriptive statistics (mean±SD) of kinematic and kinetic NHE parameters averaged for both limbs. Time and mode (assisted see above unassisted execution) effects are presented by the respective *P*-values

	NHE kinetics			NHE kinematics						DOMS
	PM_{NHE} (Nm/kg)	MM_{NHE} (Nm/kg)	J_{NHE} (Nm•s)	t_{NHE} (s)	ROM_{knee} (°)	Mean ω_{KE} (°/s)	ROM_{hip} (°)	HF_{max} (°)	ROM_{DWA} (%)	VAS
1a	3.10±0.47	2.22±0.28	15.5±2.3	7.0±0.4						3.6±1.6
2a	3.30±0.37	2.38±0.20	16.7±1.8	7.0±0.4						2.9±1.3
3a	3.50±0.33	2.49±0.24	17.2±2.1	6.9±0.5						2.1±1.1
4a	3.70±0.27	2.65±0.18	18.1±2.3	6.8±0.6						1.4±1.4
5u	4.24±0.44	2.59±0.36	9.6±3.1	3.6±0.7	67.8±7.6	19.8±3.5	16.0±5.6	19.3±9.0	55.7±19.1	2.9±1.6
6a	3.79±0.37	2.74±0.25	17.2±2.3	6.3±0.5						2.0±1.3
7u	4.36±0.45	2.80±0.39	12.1±4.0	4.2±0.9	73.1±7.9	18.4±4.2	15.4±6.0	17.3±8.0	64.2±21.9	2.1±2.1
8a	3.82±0.50	2.79±0.29	18.8±2.6	6.7±0.6	81.4±5.5	12.2±1.1	10.8±4.5	9.2±6.8	99.6±1.8	2.9±1.5
9u	4.46±0.42	2.95±0.40	13.9±4.6	4.6±1.1	73.3±8.2	17.3±5.7	15.5±5.8	15.4±7.7	70.4±25.5	2.5±1.2
10u	4.50±0.49	2.97±0.46	14.6±5.7	4.8±1.3	72.6±9.1	14.1±4.5	15.2±7.2	14.9±7.6	79.6±23.5	2.1±1.1
11u	4.48±0.37	2.97±0.42	13.8±5.7	4.5±1.3	74.3±8.7	17.8±5.5	16.4±6.9	16.8±8.8	68.2±27.3	2.4±1.4
12u	4.52±0.36	3.04±0.35	14.9±4.6	4.8±1.0	75.1±7.6	16.5±4.4	16.7±6.9	18.6±8.0	72.4±25.5	2.7±1.3
P (time)	<.001	<.001	<.001	<.001	<.001	<.001	n.s.	n.s.	<.05	<.001
P (mode)	<.001	<.001	<.001	<.001	<.05	<.05	<.05	<.05	<.001	.400

PM_{NHE}, peak moment generated during NHE; J_{NHE}, impulse (M*t) generated during NHE; MM_{NHE}, mean moment generated during NHE; t_{NHE}, time under tension during NHE; ROM_{knee}, range of motion of the knee joint; mean ω_{KE}, mean knee extension velocity; ROM_{hip}, range of motion of the hip joint; HF_{max}, maximum hip flexion angle; ROM_{DWA}, percentage of range of motion to downward acceleration in relation to ROM_{knee}

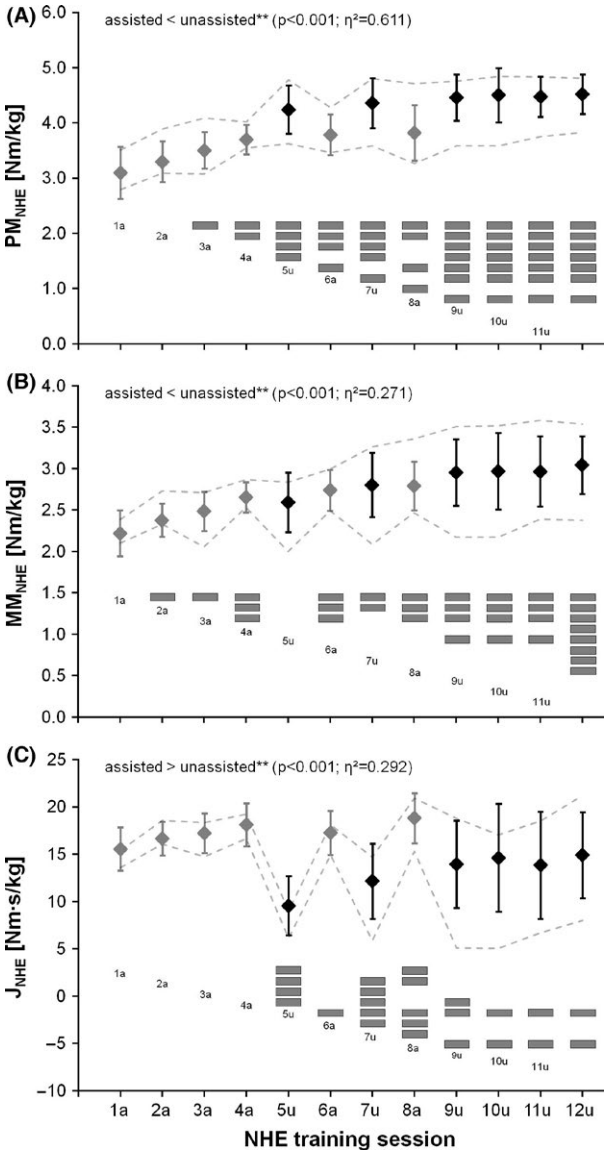

FIGURE 3 Kinetic training progress in the course of the 4-week NHE training period is illustrated as respective means and standard deviations of each of the twelve training sessions. Assisted (eg, 1a) and unassisted (eg, 5u) sessions are represented by gray and black diamonds with the main effect of NHE execution mode indicated above each graph. Dashed gray lines display the parameter time course for two exemplary participants. Significant ($P \leq .05$, gray squares) time effects are listed below each graph with reference to the respective comparative value. Peak (PM_{NHE}), mean (MM_{NHE}) moment and impulse (J_{NHE}=M*t) generated during NHE

DCRe moments (Figure 4B) displayed a comparable time course to hamstrings eccentric peak moments, but without an interaction of time and angular velocity. Besides, DCRe angles revealed a nonsignificant, but medium effect of angular velocity (Figure 4C). In contrast to all other isokinetic parameters, its significant adaptations at p1 and p2 diminish at p3 to nonsignificant increases.

4 | DISCUSSION

The discussion is structured according to the aims of the study: (a) to conduct a standardized 4-week NHE training on an isometrically operating dynamometer, (b) to include rope-assisted "isokinetic" training sessions performed with a constant knee extension velocity as well as (c) to quantify the training progress by kinetic and kinematic NHE data and (d) isokinetic parameters.

4.1 | Standardized 4-week NHE training on an isometrically operating dynamometer

Despite extensive research, standardized NHE training studies are lacking as most of them relied on partner assistance.[5,7-9,18,19,29] This procedure has two disadvantages. First, the applied resistance from hands to shanks is mostly insufficient for mature athletes to perform a NHE across the full ROM_{knee}. Beyond insufficient strength capacities of the athletes this deficient fixation results in ROM_{DWA} of below 30%.[26,27,29] Second, it is impossible to measure the generated moments quantifying training progress or fatigue. The present study was the first performing NHE on a dynamometer. In contrast to commonly conducted NHE strength tests[20-23,25] NHE kinetics was measured bilaterally which limits the applied methodology.

4.2 | Rope-assisted training sessions conducted with a constant knee extension velocity

The rationale for applying rope-assisted NHEs at a constant knee extension velocity of 15°/s was that exercise execution has up to now been based on average cadences.[26,27,29] Usually, participants are not able to adequately control their trunk's lowering speed which results in a small time under tension in the ROM near full knee extension. Within the kinematically analyzed assisted NHE session the "isokinetic" mean velocity was achieved (99.6% ROM_{DWA}) (Table 1). As hamstring fascicle length has been recently introduced to be an important factor in HSI prevention,[24,39,40] early adaptations at long fascicle length are wanted for effective preventive exercises.[6] However, due to a maximum hip flexion angle of ~10-15° most participants did not reach knee flexion angles below 30° (Figure 2). It has not yet been established if rope-assisted NHE execution is more effective in inducing increases in fascicle length compared to the unassisted one. The answer is hidden behind the mechanical determinants (eg, time under tension, impulse, peak moment, or strain rate) which all may trigger such an architectural adaptation.[39,40]

TABLE 2 Descriptive statistics (mean±SD) of isokinetic parameters obtained from knee flexors (H) and extensors (Q) averaged for both limbs. Time effects are presented by the respective *P*-values and effect sizes (η²)

	b	p0	p1	p2	p3	*P*	b	p0	p1
H			PM (Nm/kg)						APM (°)
con									
30°/s	1.66±0.21	1.65±0.17	1.70±0.17	1.70±0.18	1.75±0.18		9.7±3.1	11.0±5.3	10.1±4.9
150°/s	1.44±0.18	1.47±0.17	1.52±0.16	1.53±0.19	1.55±0.16		18.5±3.7	18.8±3.8	18.9±3.3
ecc						*P*<.001 η²= .530			
15°/s	2.06±0.26	2.04±0.24	2.17±0.26	2.18±0.28	2.19±0.29		13.1±3.9	13.5±4.1	11.9±3.3
30°/s	2.03±0.29	2.02±0.29	2.18±0.32	2.18±0.33	2.23±0.29		11.5±4.2	14.0±3.1	11.2±3.8
150°/s	2.01±0.32	2.02±0.28	2.18±0.31	2.18±0.31	2.30±0.26		16.4±5.0	14.5±3.9	14.9±7.3
Q			PM (Nm/kg)						APM (°)
con									
30°/s	3.46±0.47	3.49±0.46	3.42±0.46	3.29±0.41	3.42±0.44		73.8±5.5	73.3±6.9	73.7±6.8
150°/s	2.75±0.32	2.80±0.27	2.76±0.30	2.69±0.30	2.77±0.24	*P*=.209	70.5±6.9	72.3±6.9	73.0±5.4
ecc						η²=.098			
30°/s	4.40±0.92	4.37±0.78	4.37±0.95	4.22±0.90	4.38±0.89		74.4±6.5	76.5±5.5	78.0±4.0
150°/s	4.01±0.80	4.08±0.63	4.03±0.74	3.95±0.70	4.14±0.59		71.5±8.3	72.3±7.5	73.6±7.3
H:Q			DCRe moment (Nm/kg)						DCRe angle (°)
30°/s	1.80±0.27	1.79±0.25	1.88±0.26	1.88±0.27	1.93±0.26	*P*<.001	33.2±4.9	33.3±5.4	34.8±5.6
150°/s	1.82±0.25	1.84±0.22	1.94±0.24	1.94±0.22	1.97±0.20	η²=.538	33.7±4.0	34.0±4.7	37.6±5.5

H, hamstrings; Q, quadriceps; PM, peak moment generated during isokinetic tests; APM, angle of peak moment during isokinetic tests; CW, contractional work generated during isokinetic tests; DCRe, dynamic control ratio at the equilibrium point; ecc, eccentric; con, concentric; b, baseline; p0, pre; p1, post 1; p2, post 2; p3, post 3 (for chronological details see Figure 1)

4.3 | Quantification of training progress by kinetic and kinematic NHE data

Previous kinetic analyses of unassisted NHE execution exclusively focussed on peak moments.[20-23,25] In contrast to the mean moment or impulse, this particular parameter just reflects one point of the generated moment-time curve (Figure 2). Unassisted NHEs resulted in higher maximum and mean loads on the eccentrically working hamstrings compared to assisted ones (Figure 3A,B). However, assisted repetitions showed greater impulses (Figure 3C) due to a longer time under tension (6.8±0.6 s vs 4.4±1.1 s) (Table 1). In contrast to peak and mean moments whose training-induced increases leveled off during week 3 and 4, the generated impulse of the last unassisted NHE training session (12u) was still 21% smaller compared to the last assisted one (8a) (Table 1) indicating that the adaptation process was incomplete after 4 weeks of NHE training. This fact allows the assumption that the impulse might be the most meaningful kinetic parameter of "isokinetic" full ROM_{knee} NHE performance for distinguishing strong and weak athletes and quantifying individual progress (Figure 3C).

The recent study was the first to evaluate NHE kinematics as control variables of training stimuli. Although deviations from the target values described in the methods in terms of ROM_{knee} (81.4° vs ~90-100°) and mean knee extension velocity (12.2°/s vs 15°/s) emerged (Table 1), other training studies did not verify NHE execution quality (mean knee extension velocity, ROM_{knee}, HF_{max}, and hand position) at all.[5,7-9,18,19] Two studies captured single training sessions and identified the angle at DWA as an important parameter of NHE performance.[26,27] We calculated the ROM_{DWA} to get information about how much ROM_{knee} was accomplished at a constant velocity. In unassisted NHEs, 56%-80% ROM_{DWA} was achieved revealing no systematic adaptation effect due to large interindividual differences (68.5%±24.4%). However, these values highly exceeded previously published values (<30%).[26,27,29] This might be a result of the initially applied rope-assisted training sessions leading to strength adaptations across the full ROM_{knee}.

No study before analyzed NHE hip joint kinematics. Maximum hip flexion angle and ROM_{hip} were significantly smaller during assisted execution which increases the resultant moments by enlarging the lever arm of the center of mass to the knee joint axis (Table 1). In the course of unassisted NHE training sessions, maximum hip flexion (17.0±8.2°) and ROM_{hip} (15.9±6.3°) did not substantiate any progress. Hip flexion reduced ROM_{knee} of unassisted NHEs (72.7±8.3°) as soon as participants' heads reached the target area before approximate full knee extension was achieved (Figure 2).

p2	p3	*P*	b	p0	p1	p2	p3	*P*
					CW (J/[kg•°])			
9.4±4.4	10.4±4.1		18.8±2.1	18.8±2.0	19.8±2.0	19.9±1.9	20.5±2.5	
19.1±7.7	17.4±3.7	*P*=.459	14.8±2.4	15.3±2.4	16.0±2.1	16.3±2.4	16.1±2.1	*P*<.001
13.2±3.7	12.0±5.6	η^2=.058	25.2±2.9	25.5±3.2	26.9±3.5	26.7±3.2	27.1±3.1	η^2=.599
10.6±6.2	9.9±4.2		23.8±2.9	24.1±3.0	25.7±2.9	25.5±2.9	25.9±3.4	
16.3±7.9	13.4±1.5		23.4±2.8	23.8±2.8	26.2±3.1	25.4±2.8	26.1±2.5	
					CW (J/[kg•°])			
73.7±6.0	71.4±6.8		37.7±5.9	37.6±5.7	37.1±5.9	36.5±5.4	38.1±5.5	
72.6±6.1	69.7±7.8	*P*=.093	28.1±3.3	28.0±3.0	27.7±3.3	28.4±3.4	28.6±2.5	*P* =.469
		η^2=.122						η^2=.055
78.1±4.4	78.1±4.6		45.7±7.7	44.4±6.4	44.7±7.4	43.4±7.2	44.9±7.3	
73.7±5.7	73.7±4.7		41.7±6.8	41.4±5.4	41.8±6.1	41.0±5.6	42.5±5.2	
					H_{ecc}:Q_{con}			
35.0±5.0	34.6±5.3	*P*<.001	0.59±0.08	0.58±0.06	0.65±0.09	0.67±0.10	0.66±0.09	*P*<.001
36.9±5.1	36.5±5.4	η^2=.316	0.74±0.10	0.73±0.08	0.80±0.12	0.82±0.14	0.84±0.11	η^2=.569

4.4 | Quantification of strength adaptations by isokinetic parameters

The 4-week regular training period (b to p0) did not affect any isokinetic parameter representing a valid baseline for the following adaptations induced by NHE training (Table 2). The selective strengthening of the eccentrically working hamstrings improved the functional H:Q ratio at 30°/s (+12%) and 150°/s (+14%) (Table 2). This confirms previous findings[5,9] and might contribute to targeted HSI prevention[10-15] However, most resistance exercises (eg, squats, lunges) incorporate both muscle groups[41] and are thus not suitable to improve thigh muscle balance. This fact underlines the exceptional status of NHE as selective eccentric hamstring exercise.[3] As Coombs and Garbutt[36] suggested, purposeful knee strengthening elicited increases of DCRe moments (+7.8%) and angles (+6.3%) (Figure 4B,C, Figure 5). This effect was induced by standardized NHE training which provoked selective improvements of hamstrings eccentric peak moments and contractional work (Table 3). Hamstrings eccentric peak moments revealed a significant interaction between time and angular velocity (*P*=.007; η^2=.159) peaking at the highest angular velocity 2 weeks after NHE training ended (+14.3%) (Figure 4A). It is generally accepted that strength gains of eccentric training are more specific to contraction mode and movement velocity than those following concentric regimens.[4,40] Concerning contraction mode, the present results confirmed this aspect (Table 3). Largest increases emerging at the highest angular velocity might rely on the fact that actual velocity-dependent adaptations of hamstrings

TABLE 3 Main interaction effects of time (t), muscle group (m), contraction mode (c), and angular velocity (v) on peak moments (PM) and contractional work (CW) of hamstring (H) and quadriceps (Q) muscles obtained from concentric (con) and eccentric (ecc) isokinetic exercise

Interaction effects	PM			CW		
	P	η^2	direction	*P*	η^2	direction
t·m	.009	.263	H>Q	.013	.230	H>Q
t·c	<.001	.392	Hecc>Hcon	<.001	.358	Hecc>Hcon
t·v	.007	.159	150>30=15°/s	.551	.054	n.s.

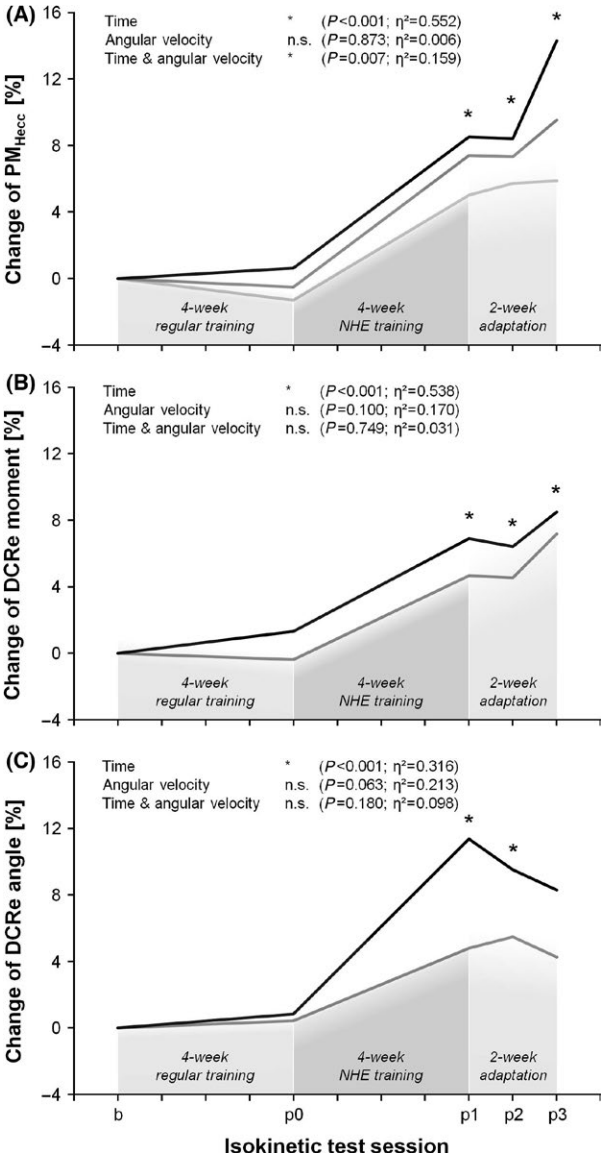

FIGURE 4 Selected isokinetic data (mean changes of pooled data) obtained at 15°/s (light gray line), 30°/s (dark gray line), and 150°/s (black line) quantify the adaptation progress of eccentric hamstrings strength (A) and muscle balance between knee flexors and extensors (B, C). Significant ($P \leq .05$) time effects related to the internal baseline period (b to p0) are indicated with *. Peak moment of isokinetic eccentric exercise (PM_{Hecc}) and thigh muscle balance parameters expressed by the dynamic control ratio at the equilibrium point (DCRe)

eccentric peak moment occurred 2 weeks after NHE training (Figure 3A). These tapering weeks might have promoted hypertrophic effects of type II muscle fiber cross-sectional area explaining the observed time course when high-velocity adaptations drifted away from slow-velocity ones (Figure 4A).[4] Additionally, it must be noted that this effect was specific for hamstrings eccentric peak moment but not for contractional work (Table 3).

Previous NHE training studies with comparable sample sizes revealed similar increases of hamstrings eccentric peak

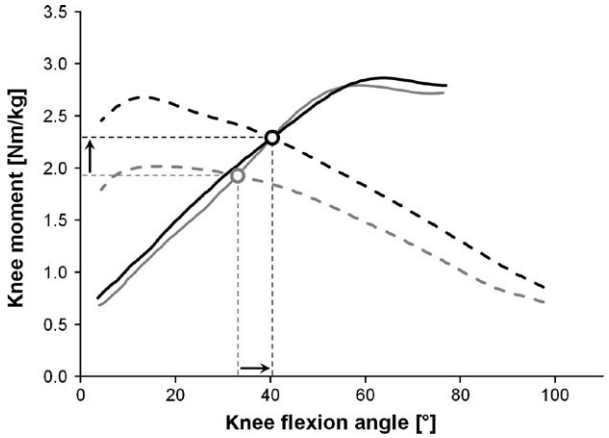

FIGURE 5 Schematic illustration of improved muscle balance (DCRe moment and angle) between concentric knee extensors (solid lines) and eccentric knee flexors (dashed lines) at 150°/s induced by NHE training comparing baseline (gray lines) with post 3 values (black lines)

moments (7%-21%).[5,7,9,19,29] All of them were conducted with partner assistance impeding individual training load quantification. As the present study comprised a short (4 weeks with 12 sessions) and low-volume NHE training program (a total of 108 NHEs) compared to the cited ones (4-10 weeks with 9-30 sessions including a total of 160-720 NHEs), the magnitude of adaptations relied neither on training duration nor volume. Instead, exercise quality, training load, and appropriate execution assistance especially at early stages (see *4.2.* and *4.3.*) might have had an impact on the training response which has not yet been reported before. The nonsignificant effect of NHE training on angles of peak moment of eccentrically working hamstrings (Table 2) confirms the results of a previous study.[7] As both studies examined participants with resistance training experience, significant changes of the shape of the hamstrings eccentric peak moment-knee flexion angle curve have not been expected (Figure 5).

4.5 | Recommendations for purposeful NHE training

Acute program variables for purposeful NHE training can hardly be derived from the present results. However, assisted NHE sessions seemed to be very useful to learn proper exercise execution and to reduce fear of uncontrolled falling especially at early training stages of inexperienced athletes. It is advisable to perform NHEs across the full ROM_{knee} right from the first repetition because the intensity of effort is important for increases of strength and hamstring fascicle length.[4,40] Assistance can be given by a rope or more easily by a partner being located in front of the athlete and adjusting the pressure of his hands to the athlete`s shoulders according to the respective movement velocity. To control the time under tension the video live feedback was very useful.

However, this feature is more appropriate for further research studies than for daily NHE training. With regard to kinetic and kinematic NHE parameters, the impulse and ROM_{DWA} appeared to best reflect the adaptations within the training period. The large standard deviations of these parameters indicated vast interindividual differences in NHE execution quality (Table 1). More importantly, appropriate tapering periods promoting post-training adaptations should be taken into account especially if strength capacities should be transferred to high-speed movements such as sprinting (Figure 4).

4.6 | Limitations

The analyzed sample consisted of a heterogenous group of sprinters substantially differing in training history, training volume, and performance level. Due to the high strength requirements of NHE, a study incorporating homogenous samples of weak and strong athletes should detect for which group NHE training is more effective. It is unknown whether exclusively rope-assisted vs unassisted NHEs or slow vs fast NHEs lead to bigger adaptations. The present results are based on bilateral NHE kinetics. Future studies conducted on dynamometers should quantify the contribution of each thigh to the generated eccentric work. The limited training duration of 4 weeks was due to participants' competitive participation. The effects of additional NHE sessions, sets and repetitions as well as retention tests warrant further investigation. The sample size of 16 participants resulted in a substantial expenditure of time (a minimum of 4-5 hours per training day) required for standardized NHE training accompanied by the same examiner. Future studies should reveal the individual relationship between training stress, isokinetic as well as NHE adaptations and distinguish between responders and nonresponders. Finally, the transferability of NHE-induced improved hamstring strength to sport-specific performance such as sprinting is of major importance.

PERSPECTIVE

The present research revealed that the NHE is an effective exercise for selective hamstring strengthening to improve muscle balance between knee flexors and extensors. This might contribute to targeted injury prevention of hamstring strain injuries [10-15] and potentially of knee injuries such as anterior cruciate ligament tears.[36,37] To our knowledge, this was the first study conducting a standardized NHE training on an isometrically operating dynamometer quantifying the training progress by kinetic and kinematic NHE data and isokinetic parameters. The present results expand the knowledge of selective eccentric hamstring training and the resultant adaptations during and after training. In the future, rising research attention should be directed toward the combined assessment of eccentric hamstring strength and muscle architecture (eg, fascicle length). This procedure will improve the valid interpretation of actual thigh muscle status by incorporating two dimensions.[24,39] However, to derive reasonable training programs, the mechanical determinants (eg, time under tension, impulse, peak moment, or strain rate) inducing the increase of fascicle length have to be determined.

ACKNOWLEDGEMENTS

There is no conflict of interest. The authors would like to thank all participants who volunteered to participate in this study and demonstrated great motivation and commitment.

REFERENCES

1. Brockett CL, Morgan DL, Proske U. Human hamstring muscles adapt to eccentric exercise by changing optimum length. *Med Sci Sports Exerc*. 2001;33:783-790.
2. Thorborg K. Why hamstring eccentrics are hamstring essentials. *Br J Sports Med*. 2012;46:463-465.
3. Ebben WP. Hamstring activation during lower body resistance training exercises. *Int J Sports Physiol Perform*. 2009;4:84-96.
4. Roig M, O'Brien K, Kirk G, et al. The effects of eccentric versus concentric resistance training on muscle strength and mass in healthy adults: a systematic review with meta-analysis. *Br J Sports Med*. 2009;43:556-568.
5. Mjolsnes R, Arnason A, Osthagen T, Raastad T, Bahr R. A 10-week randomized trial comparing eccentric vs. concentric hamstring strength training in well-trained soccer players. *Scand J Med Sci Sports*. 2004;14:311-317.
6. Guex K, Millet GP. Conceptual framework for strengthening exercises to prevent hamstring strains. *Sports Med*. 2013;43:1207-1215.
7. Iga J, Fruer CS, Deighan M, Croix MD, James DV. 'Nordic' hamstrings exercise-engagement characteristics and training responses. *Int J Sports Med*. 2012;33:1000-1004.
8. Anastasi SM, Hamzeh MA. Does the eccentric Nordic Hamstring exercise have an effect on isokinetic muscle strength imbalance and dynamic jumping performance in female rugby union players? *Isokinet Exerc Sci*. 2011;19:251-260.
9. Small K, McNaughton L, Greig M, Lovell R. Effect of timing of eccentric hamstring strengthening exercises during soccer training: implications for muscle fatigability. *J Strength Cond Res*. 2009;23:1077-1083.
10. Arnason A, Andersen TE, Holme I, Engebretsen L, Bahr R. Prevention of hamstring strains in elite soccer: an intervention study. *Scand J Med Sci Sports*. 2008;18:40-48.
11. Brooks JH, Fuller CW, Kemp SP, Reddin DB. Incidence, risk, and prevention of hamstring muscle injuries in professional rugby union. *Am J Sports Med*. 2006;34:1297-1306.
12. Gabbe BJ, Branson R, Bennell KL. A pilot randomised controlled trial of eccentric exercise to prevent hamstring injuries in community-level Australian Football. *J Sci Med Sport*. 2006;9:103-109.
13. Olsen OE, Myklebust G, Engebretsen L, Holme I, Bahr R. Exercises to prevent lower limb injuries in youth sports: cluster randomised controlled trial. *BMJ*. 2005;330:449.

14. Petersen J, Thorborg K, Nielsen MB, Budtz-Jorgensen E, Holmich P. Preventive effect of eccentric training on acute hamstring injuries in men's soccer: a cluster-randomized controlled trial. *Am J Sports Med*. 2011;39:2296-2303.
15. van der Horst N, Smits DW, Petersen J, Goedhart EA, Backx FJ. The preventive effect of the nordic hamstring exercise on hamstring injuries in amateur soccer players: a randomized controlled trial. *Am J Sports Med*. 2015;43:1316-1323.
16. Bahr R, Thorborg K, Ekstrand J. Evidence-based hamstring injury prevention is not adopted by the majority of Champions League or Norwegian Premier League football teams: the Nordic Hamstring survey. *Br J Sports Med*. 2015;49:1466-1471.
17. Goode AP, Reiman MP, Harris L, et al. Eccentric training for prevention of hamstring injuries may depend on intervention compliance: a systematic review and meta-analysis. *Br J Sports Med*. 2015;49:349-356.
18. Clark R, Bryant A, Culgan JP, Hartley B. The effects of eccentric hamstring strength training on dynamic jumping performance and isokinetic strength parameters: a pilot study on the implications for the prevention of hamstring injuries. *Phys Ther Sport*. 2005;6:67-73.
19. Tansel RB, Salci Y, Yildrim A, Kocak S, Korkusuz F. Effects of eccentric hamstring strength training on lower extremity strength of 10-12 year old male basketball players. *Isokinet Exerc Sci*. 2008;16:81-85.
20. Opar DA, Piatkowski T, Williams MD, Shield AJ. A novel device using the Nordic hamstring exercise to assess eccentric knee flexor strength: a reliability and retrospective injury study. *J Orthop Sports Phys Ther*. 2013;43:636-640.
21. Bourne MN, Opar DA, Williams MD, Shield AJ. Eccentric knee flexor strength and risk of hamstring injuries in rugby union: a prospective study. *Am J Sports Med*. 2015;43:2663-2670.
22. Opar DA, Williams MD, Timmins RG, Hickey J, Duhig SJ, Shield AJ. Eccentric hamstring strength and hamstring injury risk in Australian footballers. *Med Sci Sports Exerc*. 2015;47:857-865.
23. Opar DA, Williams MD, Timmins RG, Hickey J, Duhig SJ, Shield AJ. The effect of previous hamstring strain injuries on the change in eccentric hamstring strength during preseason training in elite Australian footballers. *Am J Sports Med*. 2015;43:377-384.
24. Timmins RG, Bourne MN, Shield AJ, Williams MD, Lorenzen C, Opar DA. Short biceps femoris fascicles and eccentric knee flexor weakness increase the risk of hamstring injury in elite football (soccer): a prospective cohort study. *Br J Sports Med* 2016;50:1524-1535.
25. Buchheit M, Cholley Y, Nagel M, Poulos N. The effect of body mass on eccentric knee flexor strength assessed with an instrumented Nordic hamstring device (Nordbord) in football players. *Int J Sports Physiol Perform*. 2016;11:721-726.
26. Ditroilo M, De Vito G, Delahunt E. Kinematic and electromyographic analysis of the Nordic Hamstring Exercise. *J Electromyogr Kinesiol*. 2013;23:1111-1118.
27. Marshall PW, Lovell R, Knox MF, Brennan SL, Siegler JC. Hamstring fatigue and muscle activation changes during six sets of nordic hamstring exercise in amateur soccer players. *J Strength Cond Res*. 2015;29:3124-3133.
28. Sconce E, Jones P, Turner E, Comfort P, Graham-Smith P. The validity of the nordic hamstring lower for a field-based assessment of eccentric hamstring strength. *J Sport Rehabil*. 2015;24:13-20.
29. Delahunt E, McGroarty M, De Vito G, Ditroilo M. Nordic hamstring exercise training alters knee joint kinematics and hamstring activation patterns in young men. *Eur J Appl Physiol*. 2016;116:663-672.
30. Alt T, Knicker AJ, Strüder HK. Factors influencing the reproducibility of isokinetic knee flexion and extension test findings. *Isokinet Exerc Sci*. 2014;22:333-342.
31. Alt T, Knicker AJ, Struder HK. The dynamic control ratio at the equilibrium point (DCRe): introducing relative and absolute reliability scores. *J Sports Sci*. 2017;35:688-693.
32. Alt T, Richert M, Schwarz YM, Knicker AJ, Strüder HK. Minimal fixation suffices for supine isokinetic knee extension: a kinetic and 3D kinematic analysis. *Isokinet Exerc Sci*. 2017;24:301-311.
33. Wilhite MR, Cohen ER, Wilhite SC. Reliability of concentric and eccentric measurements of quadriceps performance using the KIN-COM dynamometer: the effect of testing order for three different speeds. *J Orthop Sports Phys Ther*. 1992;15:175-182.
34. Parcell AC, Sawyer RD, Tricoli VA, Chinevere TD. Minimum rest period for strength recovery during a common isokinetic testing protocol. *Med Sci Sports Exerc*. 2002;34:1018-1022.
35. Aagaard P, Simonsen EB, Magnusson SP, Larsson B, Dyhre-Poulsen P. A new concept for isokinetic hamstring: quadriceps muscle strength ratio. *Am J Sports Med*. 1998;26:231-237.
36. Coombs R, Garbutt G. Developments in the use of the hamstring/quadriceps ratio for the assessment of muscle balance. *J Sports Sci Med*. 2002;1:56-62.
37. Dvir Z, Eger G, Halperin N, Shklar A. Thigh muscle activity and anterior cruciate ligament insufficiency. *Clin Biomech (Bristol, Avon).* 1989;4:87-91.
38. Cohen J. *Statistical Power Analysis For The Behavioral Sciences*, 2nd edn. Hillsdale, NJ: Erlbaum; 1988.
39. Timmins RG, Ruddy JD, Presland J, et al. Architectural changes of the biceps femoris long head after concentric or eccentric training. *Med Sci Sports Exerc*. 2016;48:499-508.
40. Timmins RG, Shield AJ, Williams MD, Lorenzen C, Opar DA. Architectural adaptations of muscle to training and injury: a narrative review outlining the contributions by fascicle length, pennation angle and muscle thickness. *Br J Sports Med* 2016;Epub ahead of print.
41. Ebben WP, Feldmann CR, Dayne A, Mitsche D, Alexander P, Knetzger KJ. Muscle activation during lower body resistance training. *Int J Sports Med*. 2009;30:1-8.

How to cite this article: Alt T, Nodler YT, Severin J, Knicker AJ, Strüder HK. Velocity-specific and time-dependent adaptations following a standardized Nordic Hamstring Exercise training. *Scand J Med Sci Sports*. 2017;00:1–12. https://doi.org/10.1111/sms.12868